

WEEK – 5 BA

Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

Description

Students must design a multi-asset financial risk simulator. They are required to compute the historical covariance matrix for 500 assets, execute a Cholesky decomposition to correlate multi-dimensional random noise, simulate independent geometric Brownian motion paths, and calculate the portfolio's 99% Value-at-Risk (VaR).

Python Implementation

```
import numpy as np

# Parameters
n_assets = 500
n_days = 252
n_simulations = 5000
confidence = 0.99

np.random.seed(42)

# Generate historical returns
historical_returns = np.random.normal(
    0.0005, 0.02, (n_days, n_assets)
)

# Historical covariance matrix
cov_matrix = np.cov(historical_returns, rowvar=False)

# Cholesky decomposition
L = np.linalg.cholesky(
    cov_matrix + np.eye(n_assets) * 1e-8
)

# Portfolio weights
weights = np.ones(n_assets) / n_assets

# Initial portfolio value
initial_value = 1000000

# Mean returns and volatility
mean_returns = historical_returns.mean(axis=0)

# Monte Carlo simulation
Z = np.random.normal(
    size=(n_simulations, n_assets)
)

# Correlated random noise
correlated_Z = Z @ L.T

# Simulated daily returns
simulated_returns = (
    mean_returns + correlated_Z
)

# Portfolio returns
portfolio_returns = (
    simulated_returns @ weights
)

# Simulated portfolio values
portfolio_values = (
    initial_value * (1 + portfolio_returns)
)
```

```
# 99% VaR
losses = initial_value - portfolio_values
VaR_99 = np.percentile(losses, 99)

print("Covariance Matrix Shape:", cov_matrix.shape)
print("Cholesky Matrix Shape:", L.shape)
print("99% Portfolio VaR: ■", round(VaR_99, 2))
```

Key Steps

1. Generate historical return data for 500 assets.
2. Calculate the 500×500 historical covariance matrix.
3. Apply Cholesky decomposition to obtain correlated random shocks.
4. Run Monte Carlo simulations for portfolio returns.
5. Calculate simulated portfolio losses.
6. Estimate the 99% Value-at-Risk from the loss distribution.

Conclusion

The simulator demonstrates how Business Analytics and quantitative techniques can be used to measure portfolio risk. The 99% VaR estimates the potential portfolio loss at the selected confidence level under the simulated market conditions.